



Real-World Aligned Walking Navigation in Outdoor Gaussian Splatting Reconstructions

by

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Dear Professor Bruenig,

In accordance with the requirements of the Degree of Master of Computer Science in the School of Electrical Engineering and Computer Science, I submit the following thesis entitled “Real-World Aligned Walking Navigation in Outdoor Gaussian Splatting Reconstructions” The thesis was performed under the supervision of Jason Weigel . I declare that the work submitted in the thesis is my own, except as acknowledged in the text and footnotes, and that it has not previously been submitted for a degree at the University of Queensland or any other institution.

Yours sincerely,
Author's Signature
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	language translation	grammar/style	planning/drafting	research/background information	content creation text	content creation visual	content creation code	feedback	
ChatGPT 5.4 01/10/25–30/02/26	✓	✓	✓	✓	✓			✓	
Gemini 3.1 pro 01/03/26–05/06/26	✓	✓	✓						
ChatGPT 5.5 01/08/25–05/06/26	✓	✓	✓	✓	✓		✓	✓	

I acknowledge that I used generative AI tools, including ChatGPT 5.4, Gemini 3.1 Pro, and ChatGPT 5.5, in completing this assessment. These tools were mainly used for language translation, grammar and style improvement, planning and drafting, research/background information, text content generation, code content generation, and feedback on writing. AI helped me explore the research topic, refine the research question, identify keywords for the literature review, and improve the language, structure, and formatting of my thesis outline. All AI-generated content was critically reviewed, verified, edited, and integrated by me. I did not use AI

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Acknowledgments

I would like to express my sincere gratitude to my supervisor, Jason Weigel, for his guidance, feedback, and support throughout this project. Jason's advice helped shape the direction of the work and improved my understanding of virtual reality, 3D reconstruction, and spatial interaction.

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Abstract

This thesis presents a semi-automatic framework for enabling continuous walking navigation in outdoor virtual reality environments reconstructed using 3D Gaussian Splatting. While Gaussian Splatting provides photorealistic visual reconstruction, it does not directly provide navigation-ready geometry, collision boundaries, or semantic information about walkable areas. To address this limitation, the proposed system combines a Gaussian Splatting reconstruction with OpenStreetMap (OSM)-derived geometry. The Gaussian Splatting model is used as the visible photorealistic layer, while OSM/Blosm geometry is used as an invisible navigation and collision layer.

The system processes mobile phone imagery or video with associated GPS metadata and converts geographic coordinates into a local East-North-Up (ENU) metric coordinate frame. COLMAP and 3D Gaussian Splatting are used to reconstruct the outdoor scene, while OSM data is imported through Blosm to generate road, path, sidewalk, and building geometry. The alignment workflow provides an initial GPS-based placement between the reconstruction and map geometry, followed by manual refinement to correct residual translation and rotation errors. In Unity, the aligned Gaussian Splatting scene is rendered as the visual environment, while the OSM/Blosm meshes remain hidden and are used to constrain user movement.

The system was demonstrated in a university campus environment. Functional testing showed that, after manual alignment, users could move continuously through the reconstructed scene while being constrained to map-defined walkable areas and prevented from entering major non-walkable regions such as buildings. The results show that external geospatial data can provide a practical navigation structure for otherwise non-interactive Gaussian Splatting scenes. However, the project also shows that fully automatic and highly accurate geographic registration remains challenging, and manual refinement is still required to obtain a usable alignment for VR navigation.

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Chapter 1

Introduction

This project investigates continuous walking navigation within large-scale outdoor 3D environments reconstructed using Gaussian Splatting. Gaussian Splatting can produce highly photorealistic scenes from image sets, but the resulting representation does not inherently provide the navigation-ready geometry, collision model, or semantic understanding required for realistic user interaction in virtual reality (VR). This limits its use in immersive applications such as virtual campus tours, urban planning visualisations, and digital heritage preservation, where believable traversal is central to the user experience [1]. Existing VR viewers for reconstructed scenes often rely on teleportation-style navigation or unconstrained movement. While teleportation can reduce VR sickness, it disrupts spatial continuity and does not replicate how people naturally explore physical environments [2].

This creates a need for a navigation framework that preserves the visual realism of Gaussian Splatting while constraining movement according to real-world spatial structure.

1.1 Background

Current approaches to creating interactive 3D scenes from images fall primarily into two categories. Traditional mesh-based photogrammetry generates explicit surface geometry that can support physics and navigation, but it may struggle to capture fine visual detail and photorealism efficiently [3]. Neural Radiance Fields (NeRF) represent scenes as continuous radiance fields for novel-view synthesis [4, 5], while 3D Gaussian Splatting uses optimized Gaussian primitives to achieve high-quality real-time rendering [6]. Although these volumetric and point-based methods can produce visually compelling results, they are not directly walkable because they do not provide an explicit navigation mesh, collision model, or semantic distinction between walkable and non-walkable areas. As a result, the interaction problem

shifts from scene reconstruction alone toward VR locomotion design, movement constraints, and spatial alignment [2, 7].

External geospatial sources provide a practical way to introduce these constraints. OpenStreetMap (OSM) supplies crowdsourced vector map data, including roads, paths, sidewalks, and building footprints [8]. Tools such as OSMnx support street-network extraction and analysis [9], while building-footprint and satellite-image alignment studies show how map and imagery features can support geographic registration [10, 11]. However, the specific problem of combining the visual fidelity of Gaussian Splatting with map-derived geometric constraints for continuous VR walking navigation remains underexplored. In particular, it is still unclear how external, non-visual geographic data can be used as an invisible navigation layer inside a primarily visual reconstruction while preserving a coherent and immersive user experience.

1.2 Aim

The overall aim of this project is to develop and evaluate a system that enables continuous, physically plausible walking navigation within large-scale, photorealistic outdoor 3D environments generated by Gaussian Splatting. The system achieves this by combining a Gaussian Splatting visual reconstruction with OpenStreetMap (OSM)-derived geometry in a shared local coordinate frame. Rather than relying on the Gaussian Splatting reconstruction itself for navigation, the system uses OSM/Blosm geometry as an invisible navigation and collision layer. The alignment process is semi-automatic: GPS metadata is used to establish an initial ENU-based alignment, and final manual adjustment is used to correct residual translation, rotation, and scale errors before the scene is used in VR. The specific aims are:

- Aim 1: To develop a practical alignment workflow between a 3D Gaussian Splatting reconstruction and OpenStreetMap (OSM)-derived geometry.

Acceptance Criterion: The system must support an initial geographic alignment by converting GPS metadata into a local ENU metric coordinate frame and aligning the reconstruction with OSM/Blosm geometry. The workflow should provide a usable coarse alignment that can be manually refined in Unity to correct residual translation, rotation, or scale errors. Alignment quality will be assessed by visual inspection and by comparing the placement of major scene features, such as building outlines, roads, and paths, against the corresponding OSM geometry.

- Aim 2: To create a continuous and physically plausible walking navigation experience in VR using manually refined OSM/Blosm geometry as an invisible navigation layer.

Acceptance Criterion: After alignment and manual adjustment, users should be

able to move continuously along OSM-defined walkable surfaces such as roads, sidewalks, and paths, while being restricted from entering non-walkable areas such as buildings. The system will be considered successful if users can complete representative wayfinding tasks without navigation-breaking failures such as falling through the world, passing through major barriers, or becoming stuck outside the walkable area.

Chapter 2

Literature review

2.1 2D/3D Data and 3D Reconstruction

The field of 3D reconstruction has evolved from traditional photogrammetric methods to learning-based scene representations [12]. Classical workflows such as Structure from Motion (SfM) and Multi-View Stereo (MVS) remain important foundations for estimating camera poses and reconstructing scene geometry from image collections [13, 14]. In outdoor environments, reconstruction quality is strongly influenced by capture strategy, viewpoint coverage, image overlap, lighting variation, and the presence of dynamic objects [15].

Neural Radiance Fields (NeRF) represent scenes implicitly by learning a mapping from spatial position and viewing direction to colour and volume density [4, 5]. NeRF-based methods can produce high-quality novel views and model complex view-dependent appearance, but their training and rendering costs can limit their use in real-time interactive applications. 3D Gaussian Splatting (3DGS) provides an alternative explicit scene representation. Instead of representing the scene as a continuous neural field, 3DGS models it as a collection of optimised 3D Gaussian primitives. Each Gaussian stores spatial and appearance properties, including position, covariance or scale and rotation, opacity, and colour information [6]. This explicit representation enables high-quality real-time rendering and is therefore well suited to interactive applications such as virtual reality (VR) [6, 15].

In a typical 3DGS pipeline, camera intrinsics and extrinsics are first estimated using an SfM system such as COLMAP. The Gaussian primitives are then optimised against the input images so that rendered views match the training views [13, 14]. The optimisation process adjusts the position, shape, opacity, and appearance of the Gaussians through differentiable rendering. Recent neural-field research also shows a broader movement toward richer scene representations that combine appearance, depth, and semantic features from sparse observations [16].

Appearance in 3DGS is not represented only by a fixed RGB value. To model view-dependent colour, 3DGS commonly uses spherical harmonics coefficients. Spherical harmonics provide a set of basis functions defined over viewing directions, allowing the colour of a Gaussian to vary depending on the direction from which it is observed. During rendering, visible Gaussians are projected onto the image plane and composited in depth order using alpha blending. A simplified front-to-back colour compositing equation is:

$$C = \sum_{i=1}^N c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j)$$

where c_i is the colour contribution of the i -th Gaussian and α_i is its opacity. This differentiable rendering formulation allows Gaussian parameters to be optimised by minimising the photometric difference between the rendered images and the ground-truth training images [6]. Common evaluation metrics for reconstruction and rendering quality include Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and rendering speed measured in frames per second (FPS) [6, 15].

Although 3DGS provides a strong visual representation, it does not directly produce the type of explicit surface geometry required for physical interaction. Unlike traditional mesh-based reconstruction, a Gaussian Splatting scene does not inherently provide a watertight mesh, collision model, navigation mesh, or semantic classification of walkable and non-walkable areas. This limitation is central to this project: while Gaussian Splatting can provide the photorealistic visual layer of an outdoor environment, additional geometric information is required before the scene can support realistic walking navigation in VR.

2.2 Geospatial Data and Alignment

To ensure that a reconstructed virtual scene corresponds to a real-world location, it must be geo-referenced or aligned with geospatial data. Geospatial data provides information about the location, shape, and structure of real-world features such as roads, paths, buildings, and terrain. Common sources of geospatial data include:

- **Global Navigation Satellite System (GNSS):** GNSS provides geographic coordinates for devices such as drones and mobile phones. It is useful for obtaining an approximate real-world position during data capture, but GNSS alone should not be treated as sufficient for precise model registration in all capture conditions. Consumer-grade GPS can be noisy, especially in urban

or campus environments where buildings, trees, and signal occlusion may reduce accuracy. Therefore, image-based or map-based geo-registration remains relevant when GNSS is unavailable, noisy, or insufficiently precise [17].

- **OpenStreetMap (OSM):** OSM is a global crowdsourced geographic information database that provides freely available vector data, including building footprints, roads, paths, sidewalks, and other map features [8]. OSM data has been widely used for urban analysis and street-network processing [9,10]. However, its quality, completeness, and currency vary across regions and feature types [18].
- **Remote sensing imagery:** Satellite and aerial imagery provide high-resolution visual information about the Earth’s surface and can serve as 2D geographic references for alignment and validation [11].

Geo-referencing aims to estimate the position, orientation, and scale of a reconstructed model within a real-world coordinate system. This can be achieved by matching common features between the reconstruction and a geographic reference, such as road intersections, building corners, or building footprints [11,17]. When corresponding points or features are available, a similarity transform can be estimated to align the reconstruction with the reference coordinate system [19]. Residual mismatch can then be reduced using point-set registration methods such as Iterative Closest Point (ICP) [20].

For outdoor image-based reconstruction, a common practical approach is to first convert latitude and longitude coordinates into a local metric coordinate frame. A local East-North-Up (ENU) coordinate system is useful because it represents positions in metres relative to a chosen geographic origin. This allows GPS positions, reconstructed camera positions, and map geometry to be compared in the same coordinate frame. However, the resulting alignment is often only an initial estimate. Errors in GPS metadata, imperfect camera reconstruction, missing or inaccurate map features, and differences between visual appearance and map geometry can all produce residual misalignment.

In the context of this project, geospatial alignment is therefore treated as a semi-automatic workflow rather than a fully automatic registration problem. GPS metadata is used to establish an initial ENU-based alignment, while OSM-derived geometry provides a real-world reference for roads, paths, sidewalks, and building boundaries. Manual refinement can then be applied to correct remaining translation, rotation, or scale errors before the scene is used for VR navigation. This approach reflects the practical difficulty of obtaining fully automatic and highly accurate alignment from consumer mobile-phone data and crowdsourced map geometry alone.

2.3 Navigation in Virtual Reality (VR)

VR navigation refers to the methods by which a user moves through and explores a virtual environment. Common navigation methods include:

- **Teleportation:** Teleportation allows the user to instantly move from one location to another. It can reduce VR sickness because it avoids continuous visual motion, but it can also reduce spatial continuity and make movement feel less like real-world traversal [2].
- **Continuous locomotion:** Continuous locomotion simulates walking through joystick input, controller input, in-place movement, or body motion. It can provide a stronger sense of spatial continuity than teleportation, but it requires careful design to manage comfort, disorientation, and motion sickness [7, 21].
- **Real walking:** Real walking maps the user’s physical walking in a tracked space into movement in the virtual world. It can provide strong spatial updating, presence, and navigation performance benefits, but it is limited by the size of the available physical space [22, 23].

The navigation approach in this project is best understood as continuous locomotion constrained by real-world map geometry. The user moves continuously through the VR scene using controller or simulator input, while the system checks the proposed movement against an invisible navigation layer derived from OSM. This differs from pure real walking, because the project does not rely on a one-to-one mapping between a large physical walking area and the virtual campus environment. It also differs from unconstrained joystick movement, because user movement is restricted to map-defined walkable areas.

Applying continuous navigation to outdoor Gaussian Splatting scenes introduces specific challenges. First, a Gaussian Splatting model is primarily a visual representation. It does not natively provide physical properties, collision boundaries, or a navigation mesh, so users cannot directly walk within it in a physically plausible way. Second, VR locomotion research shows that teleportation, continuous locomotion, and real walking involve trade-offs between comfort, presence, spatial continuity, and available physical space [7, 24]. Third, in a campus-scale environment, the virtual scene is larger than the physical space available to most users, making a purely real-walking solution impractical.

For this reason, this project separates the visible environment from the navigation structure. The Gaussian Splatting reconstruction provides the photorealistic visual scene, while OSM/Blosm geometry provides an invisible navigation and collision layer. Roads, sidewalks, and paths can be treated as walkable surfaces, while

building geometry can be used as a physical boundary. The remaining challenge is to place these two layers into a shared real-scale coordinate system so that the user appears to walk naturally through the photorealistic reconstruction while their movement is constrained by real-world geographic structure [1, 8, 9].

2.4 Summary and Gap Statement

In summary, existing work provides strong foundations for each part of the problem. Gaussian Splatting enables photorealistic real-time rendering from image sets, but does not directly provide navigation-ready geometry or semantic walkability information [6, 15]. Geospatial data sources such as OSM provide roads, paths, sidewalks, and building footprints, but they are abstract map representations rather than photorealistic VR environments [8]. VR locomotion research provides techniques for teleportation, continuous movement, and real walking, but these techniques still require appropriate collision geometry and movement constraints to support physically plausible navigation [7, 24].

This creates three main research gaps. First, outdoor Gaussian Splatting reconstructions are difficult to align perfectly with real-world map data because consumer GPS metadata, reconstruction drift, and imperfect OSM geometry can produce residual spatial errors. Second, Gaussian Splatting scenes must be augmented with external geometry before they can support walkable navigation, because the visual representation itself does not provide reliable collision or navigation surfaces. Third, there is a lack of complete workflows that connect mobile data capture, Gaussian Splatting reconstruction, geospatial alignment, and VR navigation into a single practical system.

This research addresses these gaps by developing an integrated workflow for semi-automatic alignment and map-constrained VR navigation. The system uses mobile phone imagery or video with GPS metadata to support reconstruction and initial ENU-based alignment. It imports OSM-derived geometry through BloSM to create roads, paths, sidewalks, and building meshes. After manual refinement where necessary, the Gaussian Splatting model is used as the visible photorealistic layer, while the OSM/BloSM geometry is used as an invisible navigation and collision layer. This supports the core research question: can outdoor Gaussian Splatting reconstructions be combined with known geographic data to support continuous, physically plausible walking navigation in VR environments?

Chapter 3

Methodology

This project implements a three-layer system for real-world aligned VR navigation in outdoor Gaussian Splatting reconstructions. The three layers are the data layer, the alignment layer, and the VR layer. The data layer prepares image, GPS, and reconstruction data. The alignment layer brings the Gaussian Splatting reconstruction and OSM/Blosm geometry into a shared coordinate frame. The VR layer uses the Gaussian Splatting scene as the visible environment and the OSM/Blosm geometry as an invisible navigation and collision layer.

Table 3.1: Main system components and their roles.

Layer	Component	Role
Data	GPS/GPX processing	Generates ENU-based capture metadata
Reconstruction	COLMAP + 3DGS	Produces photorealistic scene reconstruction
Alignment	ENU + manual refinement	Places scene and map geometry together
VR	Unity + Blosm colliders	Provides constrained continuous navigation

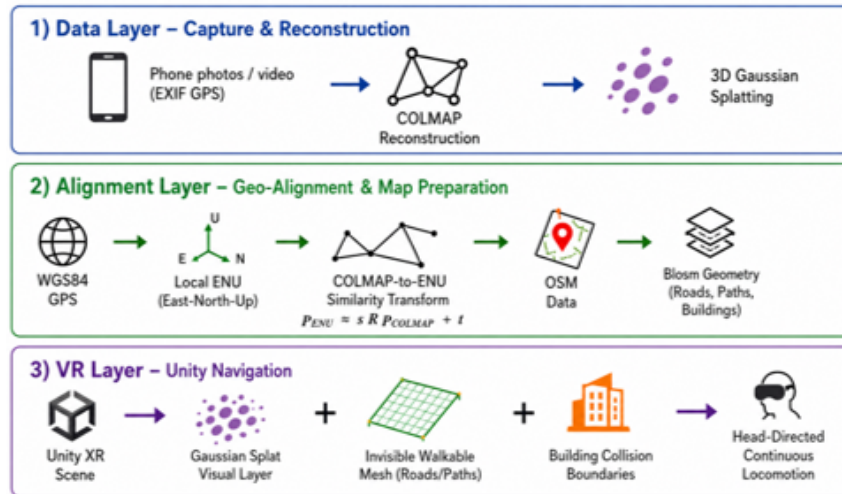


Figure 3.1: System pipeline for data processing, alignment, and VR navigation.

3.1 Data Layer

The first layer is the data and reconstruction layer. The system takes either geo-tagged phone photographs or phone video paired with a GPX track as input. The purpose of this layer is to convert raw capture data into a format that can support both 3D reconstruction and geographic alignment.

3.1.1 Data Processing and Coordinate Conversion

For photograph input, the capture pipeline extracts GPS EXIF metadata from each image. This metadata includes latitude, longitude, altitude where available, and optional compass heading. For video input, the system first extracts image frames using `ffmpeg`. Each extracted frame is then matched to an interpolated GPX position using the video start time and frame rate. This allows video data to be converted into a sequence of geotagged frames.

Both input types are converted into a unified `transforms.json` file. Each frame entry stores the image path, raw WGS84 GPS position, local ENU position, approximate heading rotation, and estimated camera intrinsics. This structure is used as the connection between the captured images, their real-world positions, and the reconstruction pipeline.

The system converts WGS84 latitude, longitude, and altitude into a local East-North-Up (ENU) coordinate system. ENU is used because it expresses positions in metres relative to a local origin. This is more suitable for reconstruction and Unity navigation than raw latitude and longitude, which are angular geographic coordinates. In this project, East and North define the horizontal navigation plane, while Up represents height. This local metric frame allows GPS camera positions, COLMAP reconstruction coordinates, and OSM/Blosm geometry to be compared and aligned.

After metadata preparation, COLMAP is used to estimate camera poses and produce the sparse reconstruction required by the 3D Gaussian Splatting pipeline. The official 3DGS training pipeline is then used to generate the Gaussian Splatting scene. The output is a photorealistic 3DGS model stored as a `.ply` file.

3.2 Alignment Layer

The second layer is the geographic alignment layer. Because COLMAP and 3DGS produce reconstructions in an arbitrary coordinate system, the reconstruction must be transformed into a real-world metric frame before it can be combined with map geometry.

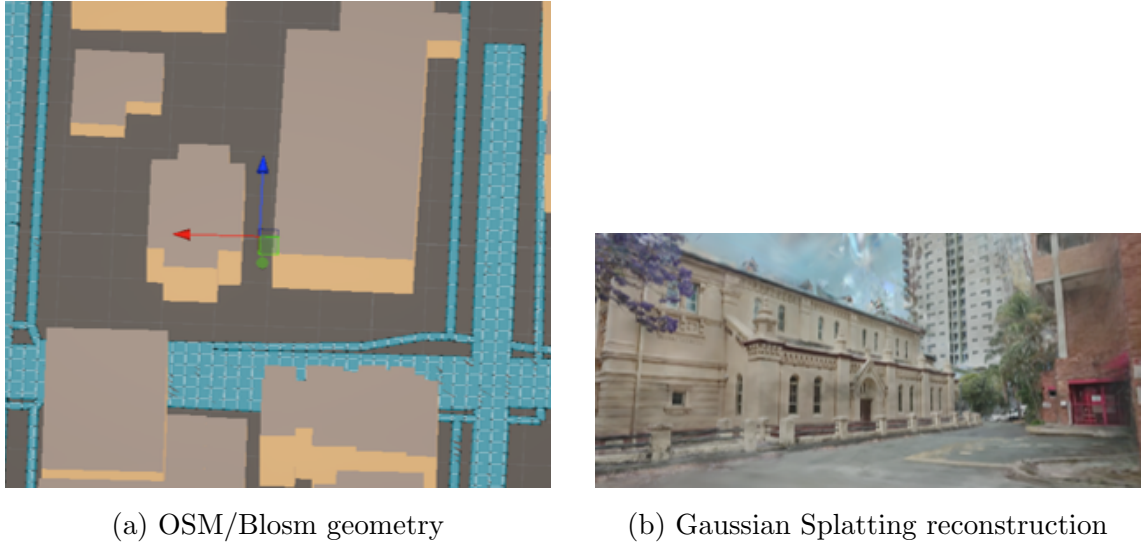


Figure 3.2: Comparison of OSM/Blosm meshes and the Gaussian Splatting reconstruction.

3.2.1 Initial ENU-Based Alignment

The system first converts GPS coordinates into the local ENU coordinate frame described above. The alignment module then matches COLMAP camera centres with their corresponding GPS-derived ENU positions from `transforms.json`. Using a Umeyama/Procrustes similarity transform, it estimates the scale, rotation, and translation required to map the COLMAP reconstruction into the ENU coordinate frame.

This similarity transform provides the initial geographic alignment. It is applied to the Gaussian Splatting `.ply` file to generate an aligned scene. However, this alignment should be understood as a coarse placement rather than a perfect automatic registration. Consumer GPS errors, reconstruction drift, and differences between visual scene content and map geometry can all introduce residual offsets.

3.2.2 OSM/Blosm Geometry Generation

OpenStreetMap is used as an external source of real-world spatial structure. The system queries OSM data around the capture area to obtain buildings and navigation-relevant features such as roads, sidewalks, footways, and paths. This map data is then imported through Blosm in Blender to generate a Unity-compatible FBX asset.

The output of this stage is a `Blosm_Map.fbx` file containing map-derived geometry. In Unity, this geometry is not intended to be the visible environment. Instead, it is used as a proxy navigation and collision layer. Meshes whose names contain terms such as `road`, `path`, `footway`, `pedestrian`, `sidewalk`, or `street` are treated as walkable surfaces. Meshes whose names contain `building` are treated as building

or collision geometry.

An optional ICP refinement step can be used to compare projected building-like splat points with OSM building footprints. In the current workflow, the visible Gaussian Splatting scene remains fixed, while the Blosm navigation geometry can be transformed using a Unity-readable `map_alignment.json` file.

3.2.3 Manual Alignment Procedure

The alignment workflow in this project is semi-automatic. The GPS/ENU conversion and similarity transform provide an initial placement between the 3DGS reconstruction and the OSM/Blosm geometry. This initial placement is useful, but it is not always accurate enough for VR navigation without further correction.

Final manual adjustment is therefore performed in Unity. During this step, the Blosm geometry or the scene alignment transform is adjusted by translation, rotation, and where necessary scale, so that major visual features in the Gaussian Splatting reconstruction correspond to the map-derived geometry. The main visual references used for this adjustment are roads, paths, sidewalks, building outlines, and other stable scene features visible in both the reconstruction and the OSM/Blosm geometry.

The alignment is considered usable when the invisible walkable surfaces are sufficiently co-located with the visible roads and paths in the 3DGS scene, and when building or non-walkable areas correspond closely enough to prevent implausible movement. This manual refinement step is important because the project does not claim fully automatic high-accuracy georegistration.

3.3 VR Layer

The third layer is the Unity VR layer. In Unity, the aligned Gaussian Splatting scene is used as the visible photorealistic environment. The Blosm-generated OSM geometry is loaded separately as an invisible navigation and collision layer. The project maps Python ENU coordinates to Unity coordinates using:

$$\text{ENU}(\text{east}, \text{north}, \text{up}) \rightarrow \text{Unity}(x, y, z) = (\text{east}, \text{up}, \text{north})$$

3.3.1 Unity Navigation Implementation

The Unity implementation separates visual rendering from movement constraints. The Gaussian Splatting reconstruction is rendered as the visual layer that the user sees. The Blosm geometry is loaded as the navigation layer, but its renderers are hidden so that it does not visually interfere with the Gaussian Splatting scene.

The `GeoAlignmentLoader` loads the Blosm FBX, applies the map alignment transform, hides the map renderers, and attaches mesh colliders to the imported geometry. The walkable meshes are used to determine whether a proposed user position is valid. Building meshes and other non-walkable geometry provide physical boundaries.

The VR navigation controller provides continuous head-directed locomotion. User input can come from XR controllers or from keyboard fallback controls in the simulator. Before applying movement, the controller checks whether the proposed position lies on the invisible Blosm walkable geometry. If the target position is inside the valid walkable area, the movement is applied. If the target position is outside the valid path area, the movement is clamped back toward the nearest valid walkable boundary.

As a result, the user visually experiences the photorealistic 3DGS reconstruction, while their movement is constrained by real-world map geometry from OpenStreetMap. This separates visual realism from navigation semantics: Gaussian Splatting provides the appearance of the environment, and OSM/Blosm provides the spatial rules that define where the user can move.

Chapter 4

Results and Discussion

4.1 Overview

This chapter presents the results of the implemented system and discusses how well the project aims were achieved. The evaluation focuses on two aspects. First, the project evaluates whether the Gaussian Splatting reconstruction, GPS metadata, and OSM/Blosm geometry can be brought into a shared local coordinate frame. Second, it evaluates whether the aligned scene can support continuous, physically plausible VR navigation using OSM/Blosm geometry as an invisible navigation and collision layer.

The system was tested using an outdoor campus environment reconstructed from mobile phone imagery. The Gaussian Splatting reconstruction was used as the visible photorealistic scene, while OSM/Blosm geometry was imported into Unity to define walkable surfaces and physical boundaries. Because the alignment process was semi-automatic, the GPS-based ENU alignment provided an initial placement, and manual adjustment was then used to correct remaining translation and rotation errors.



Figure 4.1: Alignment Result

4.2 Aim 1: Alignment Between Gaussian Splatting and OSM/Blosm Geometry

Aim 1 was to develop a practical alignment workflow between a 3D Gaussian Splatting reconstruction and OSM-derived geometry. This aim was partially achieved. The system successfully converted GPS metadata from the capture data into a local East-North-Up (ENU) metric coordinate frame. This provided a shared coordinate basis for the reconstructed camera positions, the Gaussian Splatting scene, and the OSM/Blosm map geometry.

The initial alignment produced by GPS metadata was useful as a coarse placement, but it was not sufficiently accurate to be used without further correction. This was expected because consumer mobile-phone GPS can contain several metres of error, especially in outdoor campus environments with buildings, trees, and partial signal occlusion. In addition, the Gaussian Splatting reconstruction and the OSM data do not describe the scene in the same way: the reconstruction captures visual surfaces, while OSM provides simplified vector map geometry such as building footprints, roads, paths, and sidewalks.

After the initial alignment, manual refinement in Unity was required to correct residual offsets between the visible 3DGS scene and the invisible OSM/Blosm navigation geometry. The manual adjustment mainly involved translation and rotation corrections so that major scene features, such as building edges, roads, and paths, visually corresponded with the imported Blosm geometry. Once refined, the alignment was sufficient for navigation purposes: the walkable areas defined by OSM/Blosm were positioned close enough to the visible roads and paths in the Gaussian Splatting scene for the user to move through the environment in a coherent way.

The result shows that the proposed workflow is practical, but not fully automatic. The system can establish a usable initial relationship between the reconstruction and real-world map data, but manual adjustment remains necessary. Therefore, the acceptance criterion for Aim 1 was met in terms of supporting an initial ENU-based alignment and enabling manual refinement, but the system should not be considered a complete automatic georegistration solution.

Figure 4.2 summarises the alignment outcome. The initial GPS/ENU-based alignment placed the reconstructed scene in the approximate geographic location, but it was not accurate enough for direct navigation use. The amount of manual alignment required was moderate in most cases, but it could sometimes be substantial depending on the size of the initial offset. Manual refinement was therefore required as a final registration step. This mainly involved applying a small horizontal translation, yaw rotation, and minor scale adjustment so that stable reference features, such as major road edges, path junctions, and building corners, matched

the corresponding OSM/Blosm features. The refinement was limited to global scene alignment rather than editing individual objects, and after adjustment the major roads, paths, and building outlines were sufficiently co-located for navigation and route-visualisation purposes.

The large initial error can be explained by several factors. The error can be larger in urban or built environments because of signal obstruction and multipath effects. Second, converting GPS coordinates into a local ENU frame can introduce additional offset if the origin, projection, or vertical reference is not perfectly matched. Third, the photogrammetric reconstruction itself may accumulate drift or contain scale/orientation uncertainty, particularly when image coverage is uneven. Finally, OSM/Blosm geometry is not survey-grade data; road centre lines, building footprints, and imported 3D geometry may be simplified or displaced from their real-world positions. As a result, manual alignment was necessary to correct the combined effect of GPS noise, reconstruction uncertainty, coordinate transformation error, and map-data generalisation.

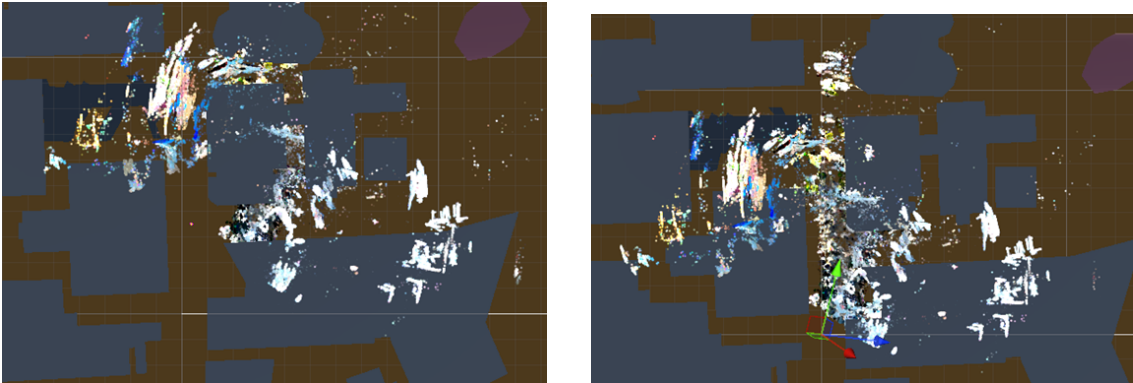


Figure 4.2: Comparison of the scene alignment before and after manual refinement.

Table 4.1: Functional navigation test cases for the VR prototype.

Test case	Expected behaviour	Result
Move along visible path	User moves continuously on walkable geometry	Passed
Move toward building area	Movement is blocked or clamped before entering building	Passed

4.3 Aim 2: Continuous Physically Plausible VR Navigation

Aim 2 was to create a continuous and physically plausible walking navigation experience in VR using manually refined OSM/Blosm geometry as an invisible navigation layer. This aim was achieved at the prototype level.

In Unity, the Gaussian Splatting reconstruction was rendered as the visible environment. The Blosm-generated OSM geometry was loaded separately and used as an invisible physical layer. Road, sidewalk, footway, pedestrian, path, and street meshes were classified as walkable surfaces, while building meshes were treated as collision or non-walkable geometry. The renderers for the Blosm geometry were disabled, so users saw the photorealistic Gaussian Splatting scene rather than the map mesh.

The navigation controller allowed continuous movement through the scene using head-directed locomotion. Before applying movement, the system checked whether the proposed user position was located on the walkable Blosm geometry. If the target position was outside the valid walkable area, the movement was clamped back toward the nearest valid point. This prevented users from freely walking through major non-walkable regions and helped maintain a physically plausible relationship between the user and the reconstructed environment.

Testing showed that, after manual alignment, users could move continuously along the main OSM-defined paths and roads within the reconstructed campus scene. The system also prevented movement into major invalid areas, such as building regions or areas outside the available walkable geometry. This demonstrates that OSM/Blosm geometry can act as an effective invisible navigation constraint for a Gaussian Splatting scene.

However, the navigation result depends strongly on the quality of the alignment and the completeness of the OSM/Blosm geometry. If the invisible navigation layer is slightly offset from the visible 3DGS scene, users may appear to be blocked before reaching a visible wall, or may be allowed to move slightly outside the visible path. Similarly, if OSM paths or sidewalks are missing or simplified, the available navigation space may not fully match the visual reconstruction. These limitations show that the system is suitable as a working prototype, but further refinement is needed for robust deployment.

4.4 Discussion

The results support the central idea of this project: Gaussian Splatting can provide the visual realism required for an immersive outdoor VR scene, while OSM/Blosm

geometry can provide the missing navigation and collision structure. This separation of responsibilities is important. The 3DGS scene is not forced to become a physical mesh, and the OSM/Blosm geometry does not need to be visually realistic. Instead, each representation is used for what it is best suited to: 3DGS for appearance, and OSM/Blosm for spatial constraints.

The main limitation is alignment accuracy. The current workflow is semi-automatic rather than fully automatic. GPS metadata provides a useful starting point, but manual correction is still required to make the visual and navigation layers coincide well enough for VR movement. This means the system is practical for a controlled demonstration environment, such as a campus scene, but would require additional automation before it could scale efficiently to many sites.

Another limitation is that OSM data is an abstraction of the real world. Roads, paths, and building footprints may be incomplete, outdated, or simplified. As a result, OSM/Blosm geometry may not perfectly match the captured scene. This affects both alignment and navigation. Nevertheless, the prototype demonstrates that even simplified map geometry can provide useful movement constraints when combined with a photorealistic Gaussian Splatting reconstruction.

Overall, the results show that the aims were met at the prototype level. Aim 1 was met by implementing a practical ENU-based alignment workflow with manual refinement. Aim 2 was met by demonstrating continuous VR navigation constrained by invisible OSM/Blosm geometry. The project therefore demonstrates the feasibility of transforming a non-interactive Gaussian Splatting reconstruction into a walkable VR environment by combining it with external geospatial data.

Chapter 5

Conclusions

5.1 Summary

This project presented a prototype workflow for enabling continuous walking navigation in outdoor virtual reality environments reconstructed using 3D Gaussian Splatting (3DGS). The main contribution of the project is the integration of three components: a photorealistic 3DGS visual reconstruction, a geospatial alignment workflow based on GPS metadata and local ENU coordinates, and an invisible OpenStreetMap (OSM)/Blosm navigation layer used to constrain user movement in Unity.

The system processes mobile phone imagery or video with associated GPS metadata and converts WGS84 latitude, longitude, and altitude into a local East-North-Up (ENU) metric coordinate frame. COLMAP and 3DGS are then used to reconstruct the outdoor scene. Since the reconstructed 3DGS model exists in an arbitrary local coordinate system, the project uses GPS-derived camera positions and similarity alignment to establish an initial relationship between the reconstruction and the real-world coordinate frame.

In parallel, OSM-derived map data is imported through Blosm to generate roads, paths, sidewalks, and building geometry. In Unity, the 3DGS reconstruction is used as the visible photorealistic layer, while the OSM/Blosm geometry is used as an invisible navigation and collision layer. User movement is controlled through a continuous locomotion controller, and proposed movement is checked against the walkable Blosm geometry. This allows the user to visually experience the Gaussian Splatting scene while being constrained by real-world map geometry.

5.2 Discussion

The results show that Gaussian Splatting is well suited for producing visually rich outdoor VR scenes, but it cannot directly support physically plausible navigation on its own. A 3DGS model does not provide a conventional mesh surface, semantic walkable regions, or collision boundaries. Therefore, additional proxy geometry is required before users can move through the scene in a controlled and realistic way.

The use of OSM/Blosm geometry provides a practical solution to this limitation. Instead of attempting to extract a navigation mesh directly from the Gaussian Splatting point representation, the system uses map-derived roads, paths, sidewalks, and buildings as the source of navigation constraints. This separation between the visible layer and the navigation layer is useful because it allows the 3DGS model to focus on visual realism, while OSM geometry provides physical and semantic structure.

However, the alignment process remains a major limitation. The project does not achieve fully automatic high-accuracy registration between the 3DGS reconstruction and OSM geometry. GPS metadata provides a useful initial alignment, but consumer GPS errors, reconstruction drift, incomplete OSM data, and differences between map geometry and visual scene content can produce residual spatial offsets. As a result, manual adjustment is still required to refine the final placement of the visual reconstruction and the invisible navigation layer.

Another limitation is that OSM geometry is an approximation of the real world. Roads, paths, and building footprints may be incomplete, outdated, or simplified. This can affect both alignment and navigation quality. In addition, the current Unity implementation demonstrates constrained continuous locomotion, but it does not yet provide a complete large-scale navigation system with robust path planning, dynamic obstacle handling, or full user-study validation.

5.3 Conclusions

In conclusion, this project demonstrates the feasibility of combining Gaussian Splatting reconstructions with external geospatial data to create a walkable VR environment. The developed workflow shows that a photorealistic but non-interactive 3DGS scene can be extended with OSM/Blosm geometry to support continuous, physically constrained navigation.

The project also highlights an important distinction: Gaussian Splatting is highly effective as a visual representation, but it should not be treated as a complete interactive world model. For VR navigation, it must be combined with additional geometric or semantic information. In this system, OSM/Blosm geometry provides

that missing structure by acting as an invisible layer for walkable surfaces and physical boundaries.

Although the current system still requires manual refinement after initial GPS-based alignment, it provides a practical foundation for real-world aligned Gaussian Splatting VR navigation. This supports the broader idea that 3DGS can be used beyond novel view synthesis, particularly when paired with external spatial data sources for interaction, navigation, and simulation.

5.4 Possible Future Work

Several directions for future work can extend this project. First, the alignment workflow could be improved through more automatic registration methods. For example, building edges, road boundaries, or other stable landmarks could be detected from the 3DGS reconstruction and matched automatically against OSM building footprints or path geometry. More robust ICP variants or feature-based registration could reduce the amount of manual adjustment required.

Second, the Unity navigation layer could be improved. The current system uses BloSM mesh geometry as an invisible collision and walkability layer. Future work could generate cleaner navigation meshes from OSM geometry, classify different surface types more reliably, and support more advanced path planning or route guidance for campus navigation.

Third, performance optimisation should be investigated for larger environments. Large-scale 3DGS scenes can be expensive to render in VR, especially on standalone headsets. Techniques such as level of detail, spatial partitioning, splat culling, streaming, and scene segmentation could help scale the system from a local campus area to larger urban environments.

Finally, a more formal user evaluation should be conducted. Future studies could compare continuous constrained walking against teleportation-based navigation, measuring task completion, navigation errors, presence, realism, comfort, and mental workload. This would provide stronger evidence for whether map-constrained continuous navigation improves the user experience in Gaussian Splatting VR environments.

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